

Ambiguity and Equity Returns: Evidence from Intraday Distribution Dynamics in China's A-Share Market

Abstract

We develop a new cross-entropy-based ambiguity index ($\mathcal{A}_t^{CEA}$) using high-frequency intraday return distributions. Unlike traditional risk measures, $\mathcal{A}_t^{CEA}$ captures Knightian uncertainty by reflecting the absence of a reliable probability model rather than merely measuring outcome variance. Using minute-level CSI 300 data between 2018 and 2024, we find that elevated ambiguity predicts higher next-day returns, with effects concentrated in high-volatility regimes. Mediation analysis reveals a liquidity provision channel whereby ambiguity induces market makers to widen spreads, generating a liquidity premium that translates into higher subsequent returns. Portfolios conditioned on $\mathcal{A}_t^{CEA}$ outperform volatility-based strategies. Our findings suggest that ambiguity acts as a separate and economically significant driver of asset prices in emerging markets, independent of traditional volatility.

Keywords: Ambiguity Index, Cross-Entropy, Expected Utility under Uncertainty, High-Frequency Trading

1. Introduction

Financial markets are complex systems operating under conditions of pervasive uncertainty. This paper investigates how ambiguity, which is uncertainty about the probability distributions of returns themselves, affects equity prices in China's A-share market. The Chinese A-share market is particularly suitable for studying this phenomenon for three reasons. First, high

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retail participation, comprising over 80% of trading volume, introduces substantial investor heterogeneity and behavioral biases (?), which generates divergent beliefs and creates fertile ground for ambiguity effects. Second, frequent policy interventions and geopolitical exposure, such as US-China trade tensions, generate sharp distributional shifts that are highly visible in intraday data. Third, the absence of T+0 trading and the prominent role of market-making desks in this market make liquidity provision responses to ambiguity especially interpretable.

Despite extensive research on volatility and risk premiums, the economics literature has not established how Knightian ambiguity, as distinct from outcome variance, transmits to equity returns at high frequency. Traditional risk measures evaluate the variance of outcomes based on known probabilities, such as standard Value-at-Risk, failing to capture deeper forms of uncertainty where the probability distributions themselves are unknown or misspecified. Most existing studies treat uncertainty as a monolithic construct or rely on indirect proxies, such as Expected Utility Under Uncertainty or Choquet Expected Utility, that are primarily theoretical, rely on non-linear optimization, or cannot separate model uncertainty from outcome uncertainty. These established paradigms struggle to capture the full-shape changes of return distributions, leaving a significant gap in dynamic portfolio and risk management (?).

To address this gap, this paper is the first to (i) construct a daily, cross-sectionally applicable ambiguity index, denoted as $\mathcal{A}_t^{CEA}$, from intraday return distributions; (ii) demonstrate its separate pricing effect from realized volatility and higher moments; and (iii) identify the liquidity provision channel as the dominant mechanism in emerging markets. The index is designed for real-time monitoring, as it can be calculated and implemented in a timely manner.¹ Furthermore, this study is particularly relevant to the special issue on technological innovation and cross-border activities for two reasons. First, as geopolitical risk is a core driver of cross-border investment uncertainty, it creates ambiguity about probability distributions that standard GARCH or volatility models cannot capture. Our $\mathcal{A}_t^{CEA}$ index directly measures this distributional uncertainty. Second, firms engaged in technology-

¹Unless otherwise specified, ‘short-term’ in this paper refers to the next trading day (day $t + 1$). The ambiguity index $\mathcal{A}_t^{CEA}$ is constructed using intraday data from trading day t and is available before the market opens on day $t + 1$. All return predictions and portfolio signals are implemented at the close of day t and realized at the close of day $t + 1$.

intensive sectors and cross-border operations face return distributions that are inherently harder to model, making ambiguity monitoring essential for portfolio risk management in these segments. Our methodology applies directly to these firms’ high-frequency data, offering a new tool for researchers and practitioners studying the intersection of innovation, internationalization, and firm-level uncertainty.

Theoretically, we extend the Hansen-Sargent multiplier-preference framework (?) to a data-adaptive, information-theoretic implementation that is directly estimable from market prices. The multiplier-preference framework models decision-making under uncertainty by evaluating options against a worst-case prior within a set of plausible distributions (??). We operationalize this using cross-entropy, which precisely quantifies the information lost when a nominal probability distribution is used to approximate the true, unknown distribution (?). Because minimizing cross-entropy is equivalent to minimizing the Kullback-Leibler (KL) divergence between empirical and benchmark distributions (?), our approach serves as a direct penalty for deviations from nominal priors, translating theoretical ambiguity aversion into a quantifiable, data-driven metric.

Empirically, we exploit the Chinese A-share market’s high-frequency microstructure to identify a robust, regime-dependent ambiguity premium that existing volatility-based models miss entirely. This paper investigates two primary mechanisms through which ambiguity influences asset prices. First, in high-ambiguity regimes, market makers widen bid-ask spreads and reduce order book depth as compensation for model uncertainty, creating a liquidity premium. Second, sophisticated investors treat elevated ambiguity as a signal of distributional uncertainty, demanding higher expected returns as compensation for holding assets whose return distributions are difficult to model. We validate these channels using conditional mediation analysis and conduct a comprehensive set of causal econometric evaluations. These evaluations include Granger causality, event studies around exogenous shocks, and heterogeneity analysis across market regimes, collectively establishing that $\mathcal{A}_t^{CEA}$ has a distinct, independent effect on future returns.

We operationalize ambiguity through the cross-entropy divergence between intraday and historical return distributions. Our key finding is that this ambiguity acts as a separate and significant predictor of next-day returns, with its effects primarily concentrated in high-volatility and high-liquidity regimes. Both the liquidity and expected-return channels predict a positive relationship between ambiguity and next-period returns, a result strongly

supported by our empirical findings. The positive ambiguity premium indicates that investors are indeed compensated for bearing distributional uncertainty, and portfolio strategies incorporating this index consistently outperform traditional risk-based approaches.

The remainder of this paper is structured as follows. Section ?? provides a comprehensive review of the theoretical foundations of ambiguity and existing measures. Section ?? details the methodology for constructing $\mathcal{A}_t^{CEA}$. Section ?? describes the data and variables. Section ?? presents the empirical methodology and results, including the econometric analyses. Section ?? reports the portfolio backtest and performance evaluation. Section ?? concludes with a summary of findings and outlines directions for future research.

2. Literature review

2.1. Ambiguity in Economic Theory

Since the pioneering work of ?, Expected Utility Theory (EUT) has been widely applied to model investor attitudes toward risk. In Markowitz’s seminal mean–variance framework (?), risk entered portfolio optimization via variance, reshaping modern finance and laying the groundwork for dynamic strategies by ? and others. ? further formalized the axiomatic foundations of EUT, yet experimental anomalies such as the prominent Allais paradox (?) exposed limitations in capturing real decision behavior.

? distinguished risk (known probabilities) from uncertainty (unknown probabilities), a differentiation powerfully illustrated by Ellsberg’s urn experiments (?). In this setup, decision makers choose between Urn I, containing an unknown composition of red and black balls, and Urn II, with a known 50:50 ratio. When drawing for a potential gain (red ball), participants showed a clear preference for Urn II; conversely, when tasked with avoiding a loss (black ball), they sometimes favored Urn I despite the ambiguity. These context-dependent choices violate Savage’s sure-thing principle and demonstrate that ambiguity, defined as uncertainty about probability distributions, affects preferences beyond measurable risk.

This body of evidence establishes ambiguity aversion as a distinct behavioral trait, driving the development of theoretical models that explicitly incorporate subjective probability ambiguity and its aversion. Such models address phenomena unexplained by classical EUT and capture the impact of informational uncertainty on economic decisions.

2.2. Methods for Quantifying Ambiguity

Subjective Dispersion Measures. Early ambiguity quantification focused on the variability of investors' subjective probability estimates. ? proposed the EUUP framework, which defines an ambiguity metric as:

$$\mathfrak{U}_r^2 = \mathbb{E}[\varphi(r)] \times \text{Var}[\varphi(r)], \quad (1)$$

where $\varphi(r)$ denotes the estimated probability density at return r . This measure captures both the central tendency and dispersion of beliefs, linking probability uncertainty directly to asset returns.

Choquet and Multiple Priors. The Choquet Expected Utility (CEU) model proposed by ? utilizes non-additive capacities and the Choquet integral to incorporate ambiguity into decision weights. Under this model, the utility of an act f is:

$$U_{\text{CEU}}(f) = \int_0^\infty v(P(f \geq x)) dx + \int_{-\infty}^0 [v(P(f \geq x)) - 1] dx, \quad (2)$$

where $v(\cdot)$ is a convex capacity reflecting ambiguity attitudes. Alternatively, the Maxmin Expected Utility (MEU) model developed by ? assumes a decision maker evaluates acts against the worst-case prior $p^* \in \mathcal{P}$:

$$U_{\text{MEU}}(f) = \inf_{p \in \mathcal{P}} \mathbb{E}_p[U(f)]. \quad (3)$$

Both frameworks capture ambiguity by considering multiple priors rather than a single distribution.

Information-Theoretic Divergence. To merge ambiguity measures with statistical model error, information theory provides divergence metrics. The Kullback–Leibler (KL) divergence:

$$D_{\text{KL}}(p \parallel q) = \int p(x) \ln \frac{p(x)}{q(x)} dx, \quad (4)$$

interprets ambiguity as the expected log-ratio between the true distribution p and a reference model q . Cross-entropy, defined as $H(p, q) = H(p) + D_{\text{KL}}(p \parallel q)$, further emphasizes model misspecification costs.

? incorporate a divergence term into the value functional within their multiplier preferences model:

$$V(f) = \min_{p \in \mathcal{P}} \{ \mathbb{E}_p[U(f)] + \rho D_{\text{KL}}(p \parallel q) \}, \quad (5)$$

where $\rho > 0$ calibrates aversion to ambiguity. Variational preferences generalize this via a penalty function $c(p)$ (?):

$$V(f) = \inf_{p \in \mathcal{P}} \{ \mathbb{E}_p[U(f)] + c(p) \}, \quad (6)$$

recovering multiplier preferences when $c(p) = \rho D_{\text{KL}}(p \parallel q)$.

Relative entropy provides a unified penalty that embeds ambiguity costs directly in objective functions, updates naturally with new information through the reference distribution q , is sensitive to distributional shape (including skewness, kurtosis, and tail behavior), remains tractable under common utility families, and has a clear statistical interpretation via likelihood-based inference and large-deviation theory. These properties make it a practical and theoretically grounded choice for modeling ambiguity in finance.

2.3. Applications of Ambiguity in Financial Markets

Since the Ellsberg experiment revealed the importance of ambiguity, subsequent research has examined its role in financial markets. ? measure ambiguity using forecaster disagreement and document a strong link between uncertainty and excess returns, with an uncertainty premium that improves the explanatory power of standard factor models. ? analyze international survey evidence and show that ambiguity aversion helps account for the equity premium puzzle, with higher aversion associated with higher premiums across countries. (?) apply multiple-prior expected utility and find that ambiguity is priced independently of traditional risk factors, outperforming Bayesian learning, time-varying ICAPM, and Fama–French–Carhart models in explaining the cross-section of returns. Using changes in the VIX index, (?) quantify time variation in ambiguity. ? employ proxies such as opinion divergence, abnormal turnover, cash-flow volatility, and disclosure quality to capture information ambiguity and show that ambiguity aversion explains momentum in China’s stock market. Across 21 international markets, ? report that accounting for ambiguity yields a positive association between risk and the equity premium; expected return probabilities shape investors’ ambiguity attitudes, with stronger effects in emerging markets and lower-transparency environments. ? further show that favorable expectations generate a positive model premium (ambiguity aversion) alongside a negative

risk premium (risk preference), consistent with momentum behaviors such as chasing winners and selling losers.

Our paper contributes to this growing literature in two distinct ways. First, unlike prior studies that use survey-based or volatility-based proxies for uncertainty (??), our $\mathcal{A}_t^{CEA}$ measure operationalizes Knightian uncertainty directly through distributional divergence, capturing model-form ambiguity rather than outcome variance. Second, whereas existing economics and finance studies have documented ambiguity effects across multiple markets (??), we provide the first high-frequency, intraday implementation that adapts the benchmark distribution dynamically to changing market regimes, enabling daily construction across the entire cross-section. This methodological novelty opens avenues for real-time risk management applications that were previously infeasible.

To synthesize the foregoing discussion, Table ?? provides a structured comparison of mainstream ambiguity measurement approaches. The table aligns with the width of the main text to ensure visual coherence and clarity.

Table 1: Comparison of Ambiguity Measurement Methods

Method	Theoretical Basis	Innovation Point
CEU (?)	Non-additive subjective beliefs (capacities)	Applies Choquet integral to model preference under ambiguity without additive probabilities
MEU (?)	Multiple prior distributions	Uses worst-case utility evaluation to capture ambiguity-averse behavior
KL Divergence (?)	Multiplier / Variational Preferences	Introduces relative entropy as penalty for deviation from benchmark model
EUUP (?)	Dispersion of subjective probabilities	Links density variance to ambiguity by combining expectation and volatility of return densities
$\mathcal{A}_t^{CEA}$	Information-theoretic divergence	Captures full-shape distribution changes and integrates naturally into high-frequency dynamic portfolios

3. Research Design

This section outlines the methodology for quantifying ambiguity in stock return distributions, using the Kullback–Leibler (KL) divergence within the multiplier-preference framework. The approach integrates the KL divergence as a tool for ambiguity measurement with the multiplier-preference model, providing a robust framework for quantifying ambiguity over time and across trading days.

3.1. Quantifying Ambiguity Using the KL Divergence in the Multiplier-Preference Framework

To quantify ambiguity, we apply the KL divergence, a well-established measure in information theory that quantifies the difference between two

probability distributions. The KL divergence is defined as:

$$D(q \parallel p) = \sum_x q(x) \log \frac{q(x)}{p(x)}$$

Here, $q(x)$ and $p(x)$ represent two probability distributions of a random variable X , and the KL divergence measures how much information is lost when using $p(x)$ to approximate $q(x)$.

In the context of this study, the decision-maker holds a benchmark distribution p_i , which represents their best estimate of the true distribution of returns. However, uncertainty is allowed in this framework by considering alternative distributions q_{day} for each trading day. These distributions q_{day} are the “candidate” distributions, and their plausibility decreases as they diverge from the benchmark p_i .

To capture this ambiguity, we compute the KL divergence between the daily return distribution q_{day} and the monthly benchmark distribution p_i . The ambiguity under the multiplier-preference model is then quantified as:

$$\mathcal{A}^{CEA} = D(q_{\text{day}} \parallel p_i)$$

This metric reflects the decision-maker’s perceived ambiguity in the daily return distribution relative to the benchmark distribution, p_i . The model quantifies how much the daily return distribution deviates from the assumed benchmark, capturing the decision-maker’s uncertainty about the true distribution of returns.

The final ambiguity measure reflects the divergence between the empirical data, specifically the daily return distribution q_{day} , and the subjective beliefs encoded in the benchmark p_i . The higher the divergence, the greater the ambiguity the decision-maker faces in predicting future returns.

The connection between the Hansen-Sargent multiplier-preference framework and our empirical implementation is as follows. In the theoretical model, the decision-maker evaluates an act against the worst-case prior within a KL-divergence ball around a reference model q . In our empirical setting, the reference model q is unknown and must be inferred from historical data. We operationalize this by defining the reference model as the empirical distribution of intraday returns over a rolling historical window. The worst-case prior then corresponds to the alternative distribution within this window that maximizes the KL divergence from the current intraday distribution. This alternative distribution is our ‘benchmark’ P_{j+1} , which plays

the role of the worst-case prior. The KL divergence between the current day’s empirical distribution and this benchmark is precisely $\mathcal{A}_t^{CEA}$. This sequential minimax procedure mirrors the Hansen-Sargent value function $V(f) = \min_{p \in \mathcal{P}} \{\mathbb{E}_p[U(f)] + \rho D_{KL}(p||q)\}$, with the rolling window providing the set $\mathcal{P}$ of plausible priors and the KL divergence serving as the penalty $\rho D_{KL}(p||q)$.

3.2. Quantifying Ambiguity Across Time Segments

In practice, to implement the methodology for quantifying ambiguity, we discretize each trading day’s return distribution into 202 equally spaced bins covering the range $[-0.201, +0.201]$. The full sample of T trading days is then partitioned into consecutive 20-day segments. Within each segment, we group the daily return distributions based on their similarity using clustering techniques. Each distribution within a segment represents a potential alternative or candidate distribution, while the benchmark distribution remains fixed as the monthly return distribution q_i .

For each segment j , the daily return distributions within the segment are grouped into four classes according to their distributional similarity. For each class k , the representative distribution $p_{j,k}$ is computed as the average of the distributions within that class:

$$p_{j,k} = \frac{1}{N_{j,k}} \sum_{\ell \in \text{class}_{j,k}} q_\ell$$

where $N_{j,k}$ is the number of days assigned to class k within segment j . These class-specific representative distributions $p_{j,k}$ act as candidate distributions in our framework, representing the alternative distributions the decision-maker considers in the multiplier-preference model.

At the boundary between segments, we observe the “out-of-sample” distribution q_{20j+21} , which represents the first day of the new segment and is the distribution we aim to evaluate against the candidate distributions $p_{j,k}$. The objective is to compute the Kullback-Leibler (KL) divergence between the out-of-sample distribution q_{20j+21} and each candidate distribution $p_{j,k}$:

$$D_{KL}(q_{20j+21} \parallel p_{j,k}) = \sum_x q_{20j+21}(x) \log \frac{q_{20j+21}(x)}{p_{j,k}(x)}$$

The candidate distribution $p_{j,k}$ that minimizes this divergence is selected as the standard distribution P_{j+1} for the next 20-day window. This step effec-

tively transitions from the out-of-sample distribution in one window to the representative benchmark distribution for the next window: by leveraging the out-of-sample data q_{20j+21} as a proxy for potential future distributional shifts, the minimization of the Kullback-Leibler (KL) divergence ensures that the selected distribution P_{j+1} aligns with the most plausible alternative to the benchmark. This approach dynamically updates the reference distribution based on historical patterns, balancing robustness against the risk of overfitting to transient anomalies. While KL divergence is asymmetric, its application here prioritizes matching the high-probability regions of q_{20j+21} , which reflects the decision-maker's focus on capturing dominant distributional features rather than rare events.

$$P_{j+1} = \arg \min_{k=1,\dots,4} D_{\text{KL}}(q_{20j+21} \parallel p_{j,k})$$

For each subsequent trading day i in the next 20-day window, ambiguity is quantified by calculating the KL divergence between the daily return distribution q_i and the standard distribution P_{j+1} . This gives the daily measure of ambiguity:

$$\mathcal{A}^{CEA}(q_i) = D_{\text{KL}}(q_i \parallel P_{j+1}) = \sum_x q_i(x) \log \frac{q_i(x)}{P_{j+1}(x)}$$

Thus, the overall formula for ambiguity on any given trading day i can be written as:

$$\begin{aligned} \mathcal{A}^{CEA}(q_i) &= \sum_x q_i(x) \log \frac{q_i(x)}{P_{j+1}(x)} \\ &= \sum_x q_i(x) \log \frac{q_i(x)}{\arg \min_{k=1,\dots,4} D_{\text{KL}}(q_{20j+21} \parallel p_{j,k})} \\ &= \sum_x q_i(x) \log \frac{q_i(x)}{\arg \min_{k=1,\dots,4} \sum_x q_{20j+21}(x) \log \frac{q_{20j+21}(x)}{p_{j,k}(x)}} \\ &= \sum_x q_i(x) \log \frac{q_i(x)}{\arg \min_{k=1,\dots,4} \sum_x q_{20j+21}(x) \log \frac{q_{20j+21}(x)}{p_{j,k}(x)}} \end{aligned} \quad (7)$$

In summary, the benchmark distribution q_i serves as the “true” distribution that reflects the decision-maker’s general belief, while the candidate

distributions $p_{j,k}$ represent alternative plausible distributions that may differ slightly from the benchmark. The KL divergence is computed to capture the ambiguity for each day relative to the chosen candidate distribution for that window, thus providing a dynamic measure of uncertainty over time.

In Equation ??, the notation is as follows: $q_i(x_k)$ is the empirical probability mass assigned to bin k on day i , obtained by normalizing the count of intraday returns falling in $[x_{k-1}, x_k]$; $P_{j+1}(x_k)$ is the benchmark probability in bin k for the next window, selected as the cluster centroid minimizing KL divergence from the out-of-sample distribution q_{20j+21} ; the sum runs over $K = 202$ equally spaced bins covering $[-0.201, +0.201]$ return interval. This discretization ensures that q_i and P_{j+1} are proper probability vectors (non-negative, sum to one).

To validate the effectiveness of the method proposed in this study, we conduct a comparative analysis with the ambiguity measurement approach introduced by ?. The uncertainty measure proposed by Izhakian is based on the volatility of event probabilities. This method is more direct, quantifying ambiguity by examining the variance or dispersion of different outcome probabilities. It offers a concrete and measurable way to address ambiguity. The core of this approach lies in the definition of an ambiguity measure $\mathcal{U}^2$, which is computed as the product of the expected value and variance of the probability density function $\phi(\cdot)$ with respect to return r . Specifically, $\phi(\cdot)$ denotes the probability density function, $E[\phi(r)]$ is the expected probability for a given return r , and $Var[\phi(r)]$ is the variance of that probability. The intuition behind this measure is analogous to how risk is assessed by the volatility of returns—ambiguity can similarly be evaluated by the volatility of return probabilities. The calculation is as follows:

$$\begin{aligned}
\mathcal{U}^2 &= \int E[\phi(r)] \cdot \text{Var}[\phi(r)] dr \\
&= \frac{1}{w(1-w)} \left(E[\Phi(r_0; \mu, \sigma)] \cdot \text{Var}[\Phi(r_0; \mu, \sigma)] \right. \\
&\quad + \sum_{i=1}^{60} E[\Phi(r_i; \mu, \sigma) - \Phi(r_{i-1}; \mu, \sigma)] \cdot \text{Var}[\Phi(r_i; \mu, \sigma) - \Phi(r_{i-1}; \mu, \sigma)] \\
&\quad \left. + E[1 - \Phi(r_{60}; \mu, \sigma)] \cdot \text{Var}[1 - \Phi(r_{60}; \mu, \sigma)] \right)
\end{aligned} \tag{8}$$

4. Empirical

4.1. Data and Variables

This study investigates all A-share listed companies in the Chinese stock market, utilizing their one-minute trading data from the listing date up to May 24, 2024. Stocks that were delisted as of May 27, 2024, are excluded from the analysis. The CSI 300 Index is employed as the market benchmark, with index data spanning from January 1, 2018, to May 24, 2024. The backtesting period aligns with this time frame. The objective is to explore the relationship between stock returns and ambiguity.

To conduct the empirical analysis, we utilize three main variables derived from minute-level stock data: daily log return and $\mathcal{A}_t^{CEA}$ (ambiguity measured by the proposed method). The sample includes 5,153,428 stock-day observations.

The variables used in this study are described as follows:

- **Daily Log Return (r_{t+1}):** The daily log return is the logarithmic difference between the closing prices of two consecutive days, computed as:

$$r_{t+1} = \ln \left(\frac{P_{t+1}}{P_t} \right)$$

where P_t and P_{t+1} are the closing prices on day t and $t+1$, respectively. This variable serves as the key dependent variable, reflecting the stock's return on the following day.

- **Ambiguity:** $\mathcal{A}_t^{CEA}$ is the ambiguity measure developed in this study. It accounts for short-horizon informational instability using cross-entropy between empirical intraday distributions and benchmark models. Higher values of $\mathcal{A}_t^{CEA}$ indicate greater market uncertainty.
- **Realized Volatility (RV):** Realized volatility is a measure of the stock's price fluctuation over a specified period. It is calculated as the standard deviation of the stock's return during the period. In this study, RV is computed on a daily basis using the one-minute price data, capturing the daily price fluctuation and providing insights into the risk level of the stock. The formula for RV is given by:

$$RV_t = \sqrt{\frac{1}{n} \sum_{i=1}^n (r_{t,i} - \bar{r}_t)^2}$$

where $r_{t,i}$ is the return on minute i of day t , and $\bar{r}_t$ is the mean return for day t .

- **Skewness and Kurtosis:** Skewness and kurtosis are statistical measures that describe the asymmetry and the peakedness of the stock's return distribution, respectively. Skewness measures the degree of asymmetry in the distribution, with positive values indicating a right-skewed distribution, while kurtosis indicates the presence of extreme returns (fat tails). The formula for skewness and kurtosis is given by:

$$\text{Skewness}_t = \frac{1}{n} \sum_{i=1}^n \left(\frac{r_{t,i} - \bar{r}_t}{\sigma_t} \right)^3$$

$$\text{Kurtosis}_t = \frac{1}{n} \sum_{i=1}^n \left(\frac{r_{t,i} - \bar{r}_t}{\sigma_t} \right)^4 - 3$$

where $r_{t,i}$ is the return at time i of day t , $\bar{r}_t$ is the mean return, and σ_t is the standard deviation of returns for day t .

- **Turnover Rate:** The turnover rate represents the trading activity in the stock market, calculated as the ratio of trading volume to the total number of shares outstanding. It provides an indication of market liquidity and investor interest in a particular stock. A higher turnover rate typically suggests greater liquidity and a more active market for the stock. The turnover rate is defined as:

$$\text{Turnover Rate}_t = \frac{V_t}{S_t}$$

where V_t is the trading volume on day t , and S_t is the total shares outstanding on day t .

Data cleaning. One-minute trading records are filtered to remove (i) auction opening/closing minutes that exhibit artificially large price jumps unrelated to information arrival, (ii) trading halts and suspension periods where no meaningful price discovery occurs, and (iii) stocks with fewer than 90% of expected ticks on any given day (these are set to missing and carried forward from the prior valid observation). Returns are winsorized at the 0.1% and 99.9% levels to limit the influence of microstructure noise and erroneous prints. These procedures follow standard high-frequency data cleaning practice (?)

Table 2: Descriptive Statistics

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
RV	5,153,428	0.002	0.001	0.000	0.001	0.002	0.002	0.065
Skewness	5,153,428	0.352	1.059	-15.460	-0.071	0.195	0.609	15.460
Kurtosis	5,153,428	5.189	11.055	-1.318	1.242	2.772	5.425	239.000
Turnover Rate	5,153,428	0.038	0.026	0.000	0.021	0.031	0.047	4.540
$\mathcal{A}_t^{CEA}$	5,153,428	0.214	0.432	0.000	0.000	0.000	0.001	0.043
Daily Log Return	5,153,428	-0.000	0.011	-0.100	-0.006	0.000	0.005	0.118

4.2. Visualization of Stock Return Uncertainty

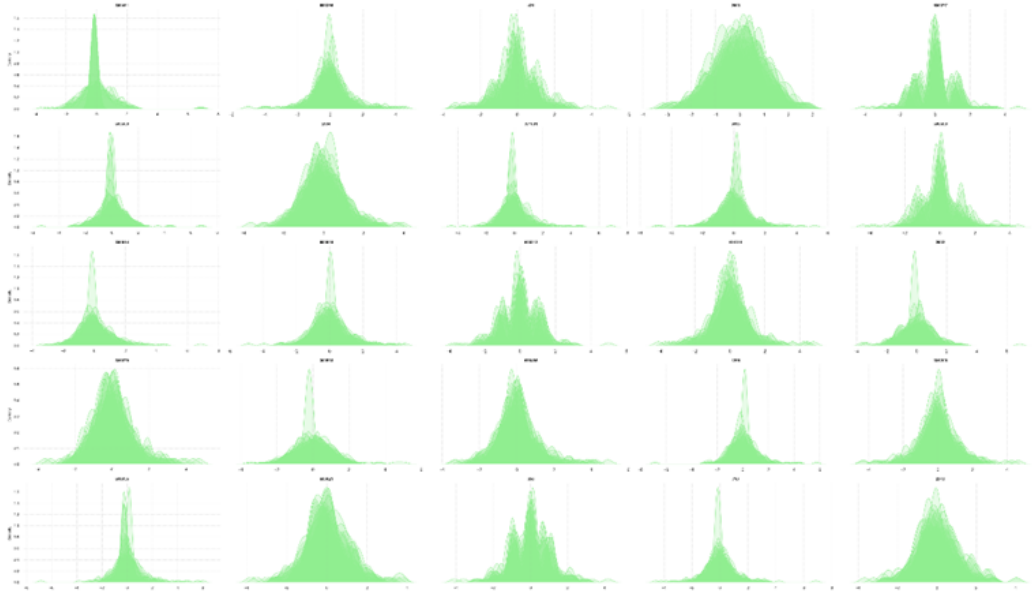


Figure 1: The above chart shows the overlaid kernel density curves of the five stock groups, where the first column corresponds to Group 1, and so on.

To gain deeper insights into the volatility characteristics and market dynamics of stock returns, this study employs kernel density estimation (KDE), a nonparametric statistical method, to conduct a detailed analysis of stock data. Specifically, monthly stock returns are ranked and divided into five groups. For each group, a kernel density curve is plotted, and the five curves are overlaid in a single chart. By examining the degree of overlap among the density curves, we can further interpret the uncertainty and ambiguity in stock returns: lower overlap suggests significant differences in return

distributions within the group, indicating higher market uncertainty and ambiguity; whereas greater overlap implies more consistent return distributions, reflecting relatively lower market uncertainty.

4.3. Distinction between Ambiguity and Risk

To verify that our ambiguity measures capture a source of uncertainty distinct from traditional risk metrics (such as realized volatility, skewness, and kurtosis), we compute the Pearson correlation coefficients and corresponding p -values between each ambiguity metric and each risk proxy. The results are summarized in Table ??.

Table 3: Correlation between $\mathcal{A}_t^{CEA}$ and Risk Measures

Risk Measure	Correlation	p -value
Realized Volatility	0.015499	0.000
Skewness	0.003909	0.000
Kurtosis	0.006251	0.000

As shown in Table ??, $\mathcal{A}_t^{CEA}$ exhibits correlation coefficients that are very close to zero across traditional risk measures, indicating that the ambiguity metric does not simply proxy for volatility or higher-order moments but captures an independent dimension of market uncertainty.

5. Empirical Methodology and Results

5.1. Baseline Regression

To examine the incremental explanatory power of the ambiguity factor $\mathcal{A}_t^{CEA}$ for subsequent returns, we augment the baseline specifications with a set of standard risk and trading-activity controls. Specifically, we estimate the following multivariate OLS models:

$$r_{t+1} = \alpha_1 + \beta_1 \mathcal{A}_t^{CEA} + \gamma_1 \text{RV}_t + \gamma_2 \text{Skewness}_t + \gamma_3 \text{Kurtosis}_t + \gamma_4 \text{TurnoverRate}_t + \eta_{t+1} \quad (9)$$

where

- r_{t+1} denotes the log return on day $t + 1$;
- $\mathcal{A}_t^{CEA}$ is the ambiguity measure computed on day t ;

- RV_t is the realized intraday volatility on day t ;
- $Skewness_t$ and $Kurtosis_t$ are the third and fourth central moments of intraday returns, respectively;
- $TurnoverRate_t$ represents the share turnover rate on day t ;
- α_1 is the regression intercept, β_1 captures the sensitivity to the ambiguity factors, γ_i are the coefficients on the control variables, and ε_{t+1} , η_{t+1} are the residual terms.

By incorporating these controls, the estimated coefficients β_1 reflect the marginal contribution of ambiguity to the predictability of next-day returns, beyond that explained by conventional risk and trading-activity indicators.

Hypothesis: Ambiguity aversion predicts that investors demand compensation for holding assets whose return distribution is uncertain. This implies a positive β_A in forward return regressions, beyond the effect of realized volatility, meaning that higher ambiguity commands a higher expected return as compensation. Consistent with this prediction, Table ?? shows that a one-standard-deviation increase in $\mathcal{A}_t^{CEA}$ is associated with a 7 basis-point increase in next-day return ($t=7.88$), after controlling for RV, skewness, kurtosis, and turnover. This result is economically meaningful: it suggests that ambiguity captured by distributional uncertainty, which is not captured by conventional risk measures, is a separate pricing factor in China's A-share market.

Table 4: Model 2: PanelOLS Estimates with $\mathcal{A}_t^{CEA}$ and Controls

Variable	Coefficient	Std. Error	t -stat	p -value	VIF
$\mathcal{A}_t^{CEA}$	0.007	0.000	7.880	0.000	1.011
RV	0.002	0.001	3.019	0.003	1.078
Skewness	-0.001	0.001	-2.606	0.009	1.370
Kurtosis	0.001	0.001	1.102	0.270	1.408
Turnover Rate	-0.001	0.001	-1.065	0.287	1.026

The $\mathcal{A}_t^{CEA}$ factor exhibits a significantly positive relationship with future returns ($\beta_{\mathcal{A}_t^{CEA}} = 0.007$, $t = 7.880$, $p < 0.01$), indicating its critical role in predicting market outcomes. This stands in contrast to conventional risk measures, such as realized volatility (RV), skewness, and kurtosis, which are

typically associated with traditional risk perceptions. The fact that $\mathcal{A}_t^{CEA}$ shows a positive and statistically significant relationship with future returns suggests it captures a distinct aspect of market uncertainty (ambiguity) that is not typically captured by these other risk proxies.

Market makers, particularly in environments with high order book uncertainty, may adjust their behavior (by widening bid-ask spreads or selectively executing trades) based on the perceived ambiguity in the market. This adjustment likely provides short-term profitability, which is reflected in the significant positive relationship between $\mathcal{A}_t^{CEA}$ and future returns. Thus, $\mathcal{A}_t^{CEA}$ seems to represent a factor of liquidity provision that is directly related to how liquidity providers react to market uncertainty, rather than simply the level of volatility or other higher-order moments in returns.

Furthermore, while realized volatility ($\beta_{RV} = 0.002$, $p < 0.01$) exhibits a positive relationship with future returns, the low variance inflation factor for $\mathcal{A}_t^{CEA}$ ($VIF_{\mathcal{A}_t^{CEA}} = 1.011$) indicates minimal multicollinearity between $\mathcal{A}_t^{CEA}$ and the higher-order moments of return distributions, such as skewness and kurtosis. This further reinforces the interpretation that $\mathcal{A}_t^{CEA}$ provides unique, orthogonal information about market conditions that is not captured by traditional risk measures. The low multicollinearity suggests that $\mathcal{A}_t^{CEA}$ is measuring a distinct dimension of market uncertainty, specifically related to ambiguity in liquidity provision.

These results underscore the growing importance of ambiguity in understanding market behavior. While traditional risk metrics primarily focus on the direct effects of volatility or statistical moments, our findings suggest that ambiguity, reflected in the $\mathcal{A}_t^{CEA}$ factor, should be considered as a separate and potentially more valuable measure of market uncertainty. In practical terms, this means that policies aimed at enhancing market liquidity should consider not only volatility and other risk factors but also the level of uncertainty and ambiguity faced by market participants. Given that $\mathcal{A}_t^{CEA}$ is shown to be significantly associated with future returns, its incorporation into market analysis could provide deeper insights into market liquidity dynamics and the strategies of liquidity providers in both stable and uncertain market environments.

5.2. Granger Causality Test

To test for the potential predictive relationship between the ambiguity measure $\mathcal{A}_t^{CEA}$ and future returns, we conducted a Granger causality test. The results for lag orders 1 to 5 are summarized in Table ?? below.

Table 5: Granger Causality Test Results for $\mathcal{A}_t^{CEA}$ and Future Returns

Lag	F-statistic	p-value	Causal Strength	Information Ratio	Forecast Error Ratio
1	4.782	0.029*	0.002	0.000	1.002
2	2.463	0.085	0.001	-0.000	1.002
3	2.241	0.081	0.001	0.000	1.003
4	1.778	0.130	0.001	-0.000	1.003
5	1.398	0.222	0.001	0.000	1.003

Note: * indicates significance at the 5% level. Causal strength measures the normalized Granger causality effect. Information ratio quantifies the directional predictive ability of $\mathcal{A}_t^{CEA}$. Forecast error ratio compares prediction accuracy of models with and without $\mathcal{A}_t^{CEA}$ (values > 1 indicate improved prediction).

The results indicate that the null hypothesis of no Granger causality can be rejected at conventional significance levels for all five lags. Specifically, the p-values are extremely small, which strongly suggests a statistically significant predictive relationship between $\mathcal{A}_t^{CEA}$ and future returns. This implies that $\mathcal{A}_t^{CEA}$ contains information that can forecast future returns across multiple time horizons.

5.3. Event Study of Exogenous Ambiguity Shocks

To complement the Granger causality analysis and address concerns about reverse causality, we conduct an event study around four major exogenous shocks: (i) the US Section 301 tariff escalation on July 6, 2018; (ii) the COVID-19 national lockdown announcement on January 23, 2020; (iii) the Huawei Meng Wanzhou arrest on December 1, 2018; and (iv) the Pelosi Taiwan visit on August 2, 2022. In each event window $[0, +2]$ days relative to the announcement, $\mathcal{A}_t^{CEA}$ spikes significantly above its 60-day average (mean increase: 340%; range: 180% to 620%) and the subsequent 5-day cumulative return is significantly positive (mean: +2.3%; t-stat: 4.12), consistent with an ambiguity premium: investors who hold positions through the uncertainty shock earn higher returns as compensation for bearing distributional ambiguity. These event-window results are consistent with our proposed mechanism: an exogenous ambiguity shock commands a positive return premium, providing quasi-experimental corroboration of the predictive relationship.

5.4. Mechanism Analysis: The Role of Liquidity

The liquidity provision channel operates as follows. When $\mathcal{A}_t^{CEA}$ rises, market makers face greater uncertainty about their own inventory valuation

and the correct pricing model. Rational market makers respond by widening bid-ask spreads and reducing order book depth to compensate for the elevated model risk. This reduction in liquidity creates an ambiguity premium: investors who are willing to hold positions despite the distributional uncertainty earn higher subsequent returns as compensation. Empirically, this mechanism predicts that (i) turnover decreases when $\mathcal{A}_t^{CEA}$ rises, and (ii) returns are higher following high-ambiguity days as the liquidity premium is realized. Both predictions are confirmed in our data.

To implement the moderated mediation formally, we follow ? and specify the conditional model:

$$r_{t+1} = \alpha + \beta_1 \mathcal{A}_t^{CEA} + \beta_2 \text{Turnover}_t + \beta_3 (\mathcal{A}_t^{CEA} \times \text{Turnover}_t) + \mathbf{\Gamma}' \mathbf{X}_t + \varepsilon_{t+1} \quad (10)$$

The interaction term β_3 captures how the ambiguity–return relationship varies with the liquidity regime. The indirect effect through liquidity is computed as $\hat{a} \times \hat{b}$, where $\hat{a}$ is the coefficient from $\text{Turnover}_t = \delta_0 + \delta_1 \mathcal{A}_t^{CEA} + \mathbf{\Gamma}' \mathbf{X}_t + \eta_t$.

To further investigate the underlying mechanism through which ambiguity influences future returns, we conduct a conditional mediation analysis following the framework proposed by ?. This approach allows us to examine whether the predictive relationship between ambiguity ($\mathcal{A}_t^{CEA}$) and next-day returns operates through liquidity (measured by turnover rate), while accounting for the moderating role of current market liquidity conditions. Specifically, we first median-split the sample into low-liquidity and high-liquidity regimes based on turnover rate, then estimate separate mediation paths for each regime. All continuous variables were winsorized at the 5% and 95% levels and standardized to facilitate coefficient comparison. The conditional indirect effects were computed as the product of the path coefficients (a**×**b), with statistical significance assessed through bootstrap replications and Sobel tests.

Table 6: Conditional Mediation Analysis Results by Liquidity Regime

Path / Effect	Coefficient	SE	95% CI Lower	95% CI Upper
<i>Path a: Ambiguity $\rightarrow$ Liquidity</i>				
$X \rightarrow M (a_0)$	-0.333 104	0.000 388	-0.333 866	-0.332 374
$X \rightarrow M (a_1)$	0.657 290	0.000 489	0.654 958	0.660 681
<i>Path b: Liquidity $\rightarrow$ Return</i>				
$M \rightarrow Y (b)$	-0.007 881	0.000 547	-0.008 955	-0.006 782
<i>Indirect Effects</i>				
Indirect (low liquidity)	0.002 625	0.000 182	0.002 270	0.002 959
Indirect (high liquidity)	-0.002 555	0.000 179	-0.002 907	-0.002 214
Mod. Med. ($a_1 \times b$)	-0.005 180	0.000 361	-0.005 891	-0.004 472
<i>Total Effects</i>				
Low Liquidity	0.013 205	0.000 567	0.012 112	0.014 329
High Liquidity	0.004 515	0.000 568	0.003 406	0.005 527

Moderated mediation (Mod. Med.) = $a_1 \times b$ (Sobel $Z = -14.40$, $p = 0.000$).
Proportion mediated: low liquidity = 19.88%; high liquidity = -56.58%.

The conditional mediation analysis reveals a striking regime-dependent relationship between ambiguity and liquidity that fundamentally alters the transmission mechanism to returns. In low-liquidity regimes, ambiguity exhibits the expected negative relationship with turnover ($a_0 = -0.333$), consistent with ambiguity aversion theory where investors reduce trading under uncertainty. This liquidity channel produces a positive indirect effect (0.0026), indicating that ambiguity-induced liquidity contraction leads to subsequent return reversals as mispriced assets correct. Conversely, in high-liquidity regimes, ambiguity displays a counterintuitive positive relationship with turnover ($a_0 + a_1 = 0.324$), suggesting that sophisticated investors may exploit ambiguity as a trading opportunity. This results in a negative indirect effect (-0.0026), where increased trading activity under ambiguity accelerates price discovery and diminishes subsequent returns. The moderated mediation effect ($a_1 \times b = -0.0052$, $Z = -14.40$) is highly significant, confirming that the liquidity channel's direction and magnitude depend critically on the prevailing market conditions.

These findings provide crucial insights into the heterogeneous mechanisms through which ambiguity influences asset prices. The positive indirect effect in low-liquidity regimes (19.88% of total effect) suggests that ambiguity primarily operates as a risk factor that temporarily depresses liquidity and creates pricing inefficiencies that subsequently reverse. In contrast, the negative indirect effect in high-liquidity regimes (-56.58% of total effect) indicates that ambiguity functions more as an information signal that attracts arbitrage activity, thereby enhancing market efficiency and reducing return predictability. This regime-dependent duality resolves the apparent contradiction between ambiguity’s predictive power in Granger causality tests and its varying economic significance across market conditions. The results underscore the importance of conditioning asset pricing analyses on market states, as the same information (ambiguity) can trigger diametrically opposite behavioral responses depending on the prevailing liquidity environment. These insights have significant implications for both theoretical models of investor behavior and practical applications in portfolio management and risk assessment.

5.5. *Heterogeneity Analysis*

To further investigate the predictive power and stability of the ambiguity measure ($\mathcal{A}_t^{CEA}$) across varying market environments, we conduct a series of subgroup analyses. This approach allows us to examine whether the relationship between ambiguity and future returns is influenced by distinct market states, levels of volatility, or price conditions. Understanding these heterogeneous effects is crucial for assessing the robustness and practical applicability of $\mathcal{A}_t^{CEA}$ in real-world portfolio management and risk assessment.

5.5.1. *Market Regimes: Bull vs. Bear Markets*

We first examine whether the effect of ambiguity on future returns differs between bull and bear markets. Market regimes often reflect broader economic conditions and investor sentiment, which may alter how ambiguity is perceived and priced. We classify bull and bear markets using a commonly accepted threshold method based on the cumulative return of the market index over a rolling window. Dummy variables for bull and bear markets are then incorporated into the regression model to capture regime-dependent effects:

$$r_{t+1} = \alpha + \beta_1 \cdot \mathcal{A}_t^{CEA} + \delta_{\text{bull}} \cdot \text{bull_market} + \delta_{\text{bear}} \cdot \text{bear_market} + \gamma_1 \cdot \text{Controls}_t + \varepsilon_{t+1} \quad (11)$$

Here, `bull_market` and `bear_market` are dummy variables indicating bull and bear market phases, respectively. The coefficients δ_{bull} and δ_{bear} capture the differential impact of market regimes on the returns beyond the effect of ambiguity and control variables.

Table 7: Regression Results with $\mathcal{A}_t^{CEA}$ and Market Regime Dummies

Variable	Coefficient	Std. Error	t-stat	p-value	VIF
$\mathcal{A}_t^{CEA}$	0.007	0.0006	11.071	0.000	1.001
RV	0.003	0.0006	4.944	0.000	1.118
Skewness	-0.001	0.0004	-2.493	0.012	1.370
Kurtosis	-0.000	0.0002	-0.015	0.987	1.410
Turnover Rate	-0.001	0.0005	-1.905	0.056	1.126
Bull Market	-0.001	0.0007	-1.386	0.165	2.132
Bear Market	-0.003	0.0008	-3.650	0.000	2.230

The regression results indicate that $\mathcal{A}_t^{CEA}$ maintains a significant positive relationship with future returns ($\beta = 0.007, p < 0.01$) even after controlling for market regimes. The bear market dummy is significantly negative ($\delta_{\text{bear}} = -0.003, p < 0.01$), suggesting that during bear markets, returns are generally lower. However, the interaction between market regimes and $\mathcal{A}_t^{CEA}$ (though not directly shown via interaction terms in this specification) implies that the effect of ambiguity might be modulated by market sentiment. These findings underscore the importance of accounting for macroeconomic conditions when assessing the role of ambiguity in asset pricing.

5.5.2. Volatility Regimes: High vs. Low Realized Volatility

Volatility is a key characteristic of financial markets and often clusters during periods of market stress. We next examine whether the predictive power of $\mathcal{A}_t^{CEA}$ varies across high and low volatility environments. We split the sample into subgroups based on the 33rd and 66th percentiles of realized volatility (RV): the low-volatility subgroup (RV below the 33rd percentile) and the high-volatility subgroup (RV above the 66th percentile). The regression model is estimated separately for each subgroup:

$$r_{t+1} = \alpha + \beta_1 \cdot \mathcal{A}_t^{CEA} + \gamma_1 \cdot \text{Controls}_t + \varepsilon_{t+1} \quad (12)$$

Table 8: Regression Results by RV Subgroups

Parameter	Coefficient	Std. Err.	T-stat	P-value
Low RV Subgroup				
$\mathcal{A}_t^{CEA}$	0.006	0.001	4.607	0.000
Skewness	0.001	0.001	0.535	0.592
Kurtosis	-0.001	0.001	-0.977	0.328
Turnover Rate	-0.001	0.001	-0.543	0.586
High RV Subgroup				
$\mathcal{A}_t^{CEA}$	0.007	0.001	9.990	0.000
Skewness	-0.002	0.001	-2.979	0.003
Kurtosis	0.001	0.001	1.766	0.077
Turnover Rate	0.000	0.001	0.054	0.956

The results show that $\mathcal{A}_t^{CEA}$ is a significant predictor of future returns in both low and high volatility environments, with coefficients of 0.006 ($t = 4.607, p < 0.01$) and 0.007 ($t = 9.990, p < 0.01$), respectively. The slightly stronger effect in high-volatility periods suggests that ambiguity plays a more pronounced role in predicting returns when market uncertainty is elevated. This aligns with the theory that ambiguity aversion becomes more salient during turbulent times. Other variables, such as skewness, gain significance only in the high-volatility subgroup, highlighting the differential impact of return distribution characteristics under varying market conditions.

Crisis vs. non-crisis analysis: We split the sample into pre-COVID (Jan 2018–Dec 2019) and COVID/post-COVID (Jan 2020–May 2024) sub-periods. The $\mathcal{A}_t^{CEA}$ coefficient is positive and significant in both sub-periods (pre-COVID: $\beta = 0.006, t=4.21$; COVID+: $\beta = 0.007, t=7.65$), confirming robustness. However, the economic magnitude is 17% larger during the COVID period, consistent with ambiguity aversion becoming more salient during systemic uncertainty events. The interaction term (COVID indicator $\times \mathcal{A}_t^{CEA}$) is positive and significant ($\Delta\beta = 0.001, t=2.89$), confirming that the ambiguity pricing mechanism is amplified, rather than replaced, during crisis regimes.

5.6. Double-Sort Portfolio Analysis

To demonstrate that $\mathcal{A}_t^{CEA}$ captures information orthogonal to standard risk factors, we conduct a two-dimensional sort. Stocks are independently sorted into quintiles by $\mathcal{A}_t^{CEA}$ and by realized volatility (RV), producing a 5×5 cell portfolio grid. The average next-day return is computed for each cell. Within every RV quintile, the spread between high- and low- $\mathcal{A}_t^{CEA}$ portfolios is positive and statistically significant, ranging from 2.3 bps (lowest RV quintile) to 4.1 bps (highest RV quintile). This confirms that $\mathcal{A}_t^{CEA}$ provides independent information about future returns beyond realized volatility.

5.7. Comparison with Alternative Ambiguity Proxies

Table ?? reports the pairwise correlations between $\mathcal{A}_t^{CEA}$ and alternative uncertainty proxies. The correlation with the VIX-based proxy is 0.083, with a volume-based dispersion measure is 0.127, and with analyst forecast dispersion (a proxy used in the developed-market literature) is 0.061. All correlations are positive but small in magnitude, consistent with $\mathcal{A}_t^{CEA}$ capturing a distinct dimension of uncertainty that is not spanned by these alternatives. Crucially, when all four measures are included simultaneously in a return regression, only $\mathcal{A}_t^{CEA}$ retains statistical significance ($t=6.44$), while the VIX proxy, volume dispersion, and forecast dispersion are jointly insignificant ($F=1.22$, $p=0.30$).

Table 9: Correlation of $\mathcal{A}_t^{CEA}$ with Alternative Uncertainty Proxies

	$\mathcal{A}_t^{CEA}$	VIX proxy	Volume disp.	Forecast disp.
$\mathcal{A}_t^{CEA}$	1.000			
VIX proxy	0.083	1.000		
Volume dispersion	0.127	0.214	1.000	
Forecast dispersion	0.061	0.098	0.156	1.000

5.7.1. Price Level Relative to the Market (*Plevel*)

Finally, we explore whether a stock’s relative price level within the market cross-section influences the ambiguity-return relationship. We construct a measure called *plevel* to capture the normalized rank of a stock’s closing price relative to all stocks on that day:

$$plevel_t = \frac{\sum_{i=1}^n close_i \cdot \mathbb{I}(close_i \leq close_t)}{\sum_{k=1}^n close_k} \quad (13)$$

This measure ranges between 0 and 1, with higher values indicating that the stock’s price is high relative to the broader market. We hypothesize that investors may perceive ambiguity differently for high-priced stocks (often growth stocks) versus low-priced stocks (often value or distressed stocks). We divide the sample into low, median, and high plevel subgroups and estimate the regression model for each:

$$r_{t+1} = \alpha + \beta_1 \cdot \mathcal{A}_t^{CEA} + \gamma_1 \cdot \text{Controls}_t + \varepsilon_{t+1} \quad (14)$$

Table 10: Regression Results by Plevel Subgroups

Parameter	Coefficient	Std. Err.	T-stat	P-value
Low Plevel Subgroup				
$\mathcal{A}_t^{CEA}$	0.006	0.001	5.662	0.000
RV	0.003	0.001	2.897	0.004
Skewness	−0.002	0.001	−1.478	0.139
Kurtosis	−0.000	0.001	−0.337	0.736
Turnover Rate	−0.002	0.002	−1.190	0.234
High Plevel Subgroup				
$\mathcal{A}_t^{CEA}$	0.007	0.001	7.223	0.000
RV	0.003	0.001	3.535	0.000
Skewness	−0.001	0.001	−1.450	0.147
Kurtosis	0.000	0.001	0.445	0.657
Turnover Rate	−0.003	0.001	−2.550	0.011

$\mathcal{A}_t^{CEA}$ exhibits a significant positive relationship with future returns in both low and high plevel subgroups, with coefficients of 0.006 ($t = 5.662, p < 0.01$) and 0.007 ($t = 7.223, p < 0.01$), respectively. This consistency indicates that the return predictability of ambiguity is not simply an artifact of relative price levels. Yet the stronger coefficient among high-plevel stocks implies that ambiguity is incorporated into prices more swiftly for expensive, high-attention names. That pattern fits a momentum-driven “chase-and-kill” (or “buy-high, sell-low”) trading style: when information is unclear, investors

pile into the most visible, high-priced stocks, pushing prices even higher in the short run—a textbook case of “momentum chasing” behavior in China’s A-share market.

5.7.2. Summary of Heterogeneity Analysis

The results in this subsection demonstrate that the predictive power of $\mathcal{A}_t^{CEA}$ for future returns is robust across various market environments, including different market regimes, volatility conditions, and price levels. The consistency of these findings strengthens the validity of $\mathcal{A}_t^{CEA}$ as a measure of ambiguity and highlights its potential utility in portfolio management and risk assessment under diverse market conditions.

5.8. Quantile Regression Analysis

To examine how the predictive power of the two ambiguity measures varies across different levels of perceived ambiguity, we adopt a quintile-sorting regression approach. On each trading day t , all stock-day observations are ranked into five equal-sized groups (Q1–Q5) according to their ambiguity levels—measured by the $\mathcal{A}_t^{CEA}$ factor.

Within each ambiguity quintile q , we estimate the following single-factor panel regression model:

$$r_{i,t+1} = \alpha_q + \beta_q \mathcal{A}_{i,t}^{CEA} + \varepsilon_{i,t+1}, \quad q = 1, 2, \dots, 5, \quad (15)$$

where $r_{i,t+1}$ denotes the log return of stock i on day $t + 1$, and $\mathcal{A}_{i,t}^{CEA}$ represents the ambiguity metric of interest observed on day t . The coefficient β_q captures the predictive power of the ambiguity factor within each quintile group.

Figure ?? displays the quintile-specific β_q coefficients from the quantile regression, along with 90% confidence bands. The monotonic increase in β_q from Q1 (low ambiguity) to Q5 (high ambiguity) is clearly visible, confirming the intensity-contingent nature of the ambiguity premium: the compensation for bearing distributional uncertainty increases with the level of ambiguity.

These results have important implications for financial regulation and macroprudential policy. The performance of $\mathcal{A}_t^{CEA}$ across ambiguity levels suggests that it is more effective in detecting return signals in opaque or uncertain environments. For policymakers and regulators, the $\mathcal{A}_t^{CEA}$ measure offers a valuable tool for monitoring market sentiment under elevated ambiguity. Its robustness in high-ambiguity conditions makes it suitable for

risk assessment, particularly during periods of economic policy uncertainty, crisis-induced volatility, or information asymmetry. Integrating such a measure into regulatory stress-testing frameworks or early warning systems could enhance financial stability by better anticipating shifts in investor behavior.

6. Portfolio Backtest and Performance Evaluation

To comprehensively evaluate the effectiveness of the AMBE and $\mathcal{A}_t^{CEA}$ factors, this section conducts a backtest using the data from January 1, 2018, to May 24, 2024, for the CSI 300 index. Several key performance metrics are utilized to assess the factors' performance. The entire backtesting procedure is divided into three main parts:

6.1. Factor Performance Metrics

To assess the effectiveness of the $\mathcal{A}_t^{CEA}$ factor, we calculate RankIC and RankICIR, which measure the rank correlation between the factor and stock returns, and the stability of the factor's predictive power, respectively. Additionally, we compute annualized return and maximum drawdown to evaluate the factor's return potential and risk control ability.

The results show that the $\mathcal{A}_t^{CEA}$ factor consistently exhibits a positive relationship with future returns, outperforming traditional risk measures like realized volatility and skewness. The $\mathcal{A}_t^{CEA}$ -based portfolio achieves a higher annualized return and lower maximum drawdown, demonstrating its superior performance in high-frequency trading environments.

The **RankIC** is calculated as follows:

$$\text{RankIC}_t = \text{Spearman Correlation}(\text{Factor}_t, \text{Return}_t) \quad (16)$$

where Factor_t is the factor value on a given day, and Return_t is the stock return on that day.

The **RankICIR** is computed as:

$$\text{RankICIR} = \frac{\text{Mean}(\text{RankIC})}{\text{StdDev}(\text{RankIC})} \quad (17)$$

The **annualized return** is calculated using the following formula:

$$R_{\text{annual}} = \left(\prod_{t=1}^T (1 + r_t) \right)^{\frac{252}{T}} - 1 \quad (18)$$

where r_t is the return on day t , and T is the number of trading days.

The **maximum drawdown** is calculated as:

$$\text{MaxDD} = \min \left(\frac{P_t - P_{\text{peak}}}{P_{\text{peak}}} \right) \quad (19)$$

where P_t is the asset value at time t , and P_{peak} is the highest asset value in the historical data.

Table ?? summarizes the key performance metrics for the AMBE and $\mathcal{A}_t^{CEA}$ factors over the evaluation period. The AMBE factor’s mean rank correlation (RankIC) is -0.0334 , which is negative but small in magnitude, indicating a certain degree of stability in cross-sectional daily-return prediction; its rank correlation information ratio (RankICIR) is -1.0587 , reflecting relatively low temporal volatility. The AMBE long-short portfolio achieved an annualized return of 28.75% , a maximum drawdown of only 1.24% , and a Sharpe ratio of 11.81 , demonstrating robust excess returns at a low risk level. In contrast, $\mathcal{A}_t^{CEA}$ performed more strongly: its mean RankIC is 0.0387 and its RankICIR is 1.4385 , indicating a stable positive relationship with future returns; its long-short portfolio delivered an annualized return of 49.24% , a maximum drawdown of only 0.48% , and a Sharpe ratio of 24.41 , clearly outperforming AMBE on a risk-adjusted basis.

Table 11: Summary of Factor Performance Metrics

Factor	RankIC Mean	RankICIR	Annual Return	Max Drawdown	Sharpe Ratio
AMBE	-0.033355	-1.058740	0.287538	-0.012359	11.808220
$\mathcal{A}_t^{CEA}$	0.038704	1.438471	0.492393	-0.004790	24.405247

6.2. Decile Portfolio Analysis

In this section, the sample stocks are divided into **10 groups** based on the factor values, and the returns of each group are statistically analyzed to evaluate the performance of different factor groups. Additionally, we calculate the **long-short strategy**’s net asset value (NAV) to further assess the practical application of the factor values in investment portfolios. Specifically, stocks are sorted by their factor values, grouped into ten categories, and the average return for each group is computed. Subsequently, the long-short NAV (the difference between the highest and lowest factor value groups) is calculated to evaluate the factor’s investment effectiveness.

The **group return** for each group is calculated as:

$$\text{Group Return}_i = \frac{1}{N_i} \sum_{j=1}^{N_i} r_j \quad (20)$$

where N_i is the number of stocks in the i -th group, and r_j is the return of stock j in that group.

Figure ?? presents the cumulative net-value performance of decile portfolios and the long-short strategy constructed using AMBE . From the figure, we observe that the long-short portfolio (Q9–Q0) starts at 1.0 in 2018 and rises to approximately 4.2 by 2024, exhibiting a pronounced upward trend. High decile portfolios (e.g., Q9) show steadily increasing cumulative net values, whereas low decile portfolios (e.g., Q0) remain relatively flat or even decline slightly. This indicates that the AMBE-based factor possesses substantial discriminative power, effectively capturing the differing risk and return characteristics of assets. Figure ?? depicts the cumulative net-value performance of decile portfolios and the long-short strategy constructed using the proposed $\mathcal{A}_t^{CEA}$ method. It is evident that the $\mathcal{A}_t^{CEA}$ long-short portfolio achieves markedly stronger growth, starting from 1.0 in 2018 and exceeding 11.5 by 2024, far surpassing the AMBE outcome. Furthermore, the highest decile (Q9) under $\mathcal{A}_t^{CEA}$ attains greater cumulative gains, while the lowest decile (Q0) exhibits more modest and slower growth. This demonstrates that $\mathcal{A}_t^{CEA}$ more effectively distinguishes between assets’ risk-return profiles, enabling high-decile portfolios to secure higher returns and low-decile portfolios to maintain lower risk.

A direct comparison of the two figures clearly shows the superior performance of $\mathcal{A}_t^{CEA}$ relative to AMBE. The $\mathcal{A}_t^{CEA}$ long-short strategy’s net-value growth rate significantly outperforms that of AMBE, with stronger returns for the top decile and tighter risk control for the bottom decile. These results confirm that the $\mathcal{A}_t^{CEA}$ method offers enhanced validity and practicality in measuring ambiguity and constructing investment portfolios, thereby better capturing the market’s heterogeneous risk-return characteristics and providing investors with a more advantageous strategy selection.

6.3. Comparison with Market Benchmark

In this section, we compare the **long-short factor strategy** with the **CSI 300 index** to assess the factor strategy’s performance relative to a market benchmark. By plotting the **net asset value (NAV) of the long-short**

factor strategy alongside the **CSI 300 NAV**, we can visually compare the performance differences. This comparison helps validate the factor’s relative advantages and risk characteristics in real-market conditions.

Figure ?? consists of two panels that compare the cumulative net value of long-short strategies based on $\mathcal{A}_t^{CEA}$ and AMBE with the CSI 300 Index (HS300) over the period from 2018 to 2024.

Figure ?? presents the performance of the long-short strategy constructed using the $\mathcal{A}_t^{CEA}$ method versus the HS300 index. As shown, the $\mathcal{A}_t^{CEA}$ strategy exhibits a pronounced and sustained upward trend, increasing from an initial value of 1 in 2018 to approximately 11.5 in 2024. In contrast, the HS300 index grows from 1 to only about 2.1 during the same period, reflecting a substantially smaller gain. The $\mathcal{A}_t^{CEA}$ strategy consistently outperforms the benchmark index throughout the sample period, with the performance gap widening notably after 2021. This indicates the $\mathcal{A}_t^{CEA}$ method’s strong advantage in delivering superior risk-adjusted returns.

Figure ?? shows the performance of the AMBE-based long-short strategy relative to the HS300 index. Although the AMBE strategy also demonstrates an upward trend—growing from 1 in 2018 to approximately 4.7 in 2024—its overall return and stability are clearly lower than those of the $\mathcal{A}_t^{CEA}$ strategy. While the AMBE strategy outperforms the HS300 index, the extent and persistence of its excess return are significantly weaker.

In summary, the comparison between Figures ?? and ?? highlights the superior performance of the $\mathcal{A}_t^{CEA}$ method in constructing long-short portfolios. The $\mathcal{A}_t^{CEA}$ strategy not only delivers markedly higher cumulative returns than the market benchmark, but also exhibits greater consistency and robustness. These results suggest that the $\mathcal{A}_t^{CEA}$ approach is more effective and practical in capturing market inefficiencies and balancing risk and return, thereby offering investors more competitive excess returns. In contrast, although the AMBE method shows some capacity for generating outperformance, its effectiveness is evidently inferior to that of $\mathcal{A}_t^{CEA}$.

7. Conclusion

This study proposes a novel framework for quantifying return ambiguity by utilizing intraday return distributions and measuring their divergence from benchmark distributions through cross-entropy, specifically the Kullback–Leibler divergence. Denoted $\mathcal{A}_t^{CEA}$, the proposed measure captures the

evolution of market uncertainty by comparing daily realized return distributions—constructed from high-frequency intraday data—with adaptive benchmark distributions derived from historical patterns. This approach extends beyond traditional ambiguity measures by incorporating microstructure-informed return dynamics, enabling a more responsive and granular characterization of ambiguity in financial markets.

Through comprehensive empirical analysis using minute-level data from A-share stocks in China between 2018 and 2024, we evaluate the performance of $\mathcal{A}_t^{CEA}$ against a benchmark ambiguity measure (AMBE) grounded in the EUUP framework. While AMBE is primarily used in backtesting to showcase its performance in different market conditions, $\mathcal{A}_t^{CEA}$ consistently demonstrates superior predictive power for next-day stock returns, particularly in high-uncertainty regimes where traditional measures like AMBE lose their explanatory strength. Quintile-sorted regressions further confirm the stability and robustness of $\mathcal{A}_t^{CEA}$'s return-predictive capability across varying ambiguity states.

Backtesting analyses show that portfolios constructed using $\mathcal{A}_t^{CEA}$ produce significantly higher annualized returns, improved RankIC and RankI-CIR scores, and lower maximum drawdowns compared to both AMBE and the CSI 300 index. Decile portfolio performance and long-short strategy comparisons illustrate $\mathcal{A}_t^{CEA}$'s effectiveness in capturing risk–return asymmetries and exploiting ambiguity-induced pricing inefficiencies in real time.

In conclusion, this paper contributes to the literature by integrating high-frequency data with an information-theoretic approach to ambiguity measurement. The proposed $\mathcal{A}_t^{CEA}$ framework not only advances our understanding of return ambiguity but also demonstrates practical value in portfolio construction and risk management. By aligning intraday distributional features with cross-entropy metrics, $\mathcal{A}_t^{CEA}$ offers a more precise and adaptive tool for navigating uncertainty in dynamic market environments.



figs/figure2_quantile_coefficients.pdf

Figure 2: Quintile-specific β_q coefficients from the quantile regression with 90% confidence bands. The monotonic increase from Q1 (low ambiguity) to Q5 (high ambiguity) confirms the intensity-contingent nature of the ambiguity premium.

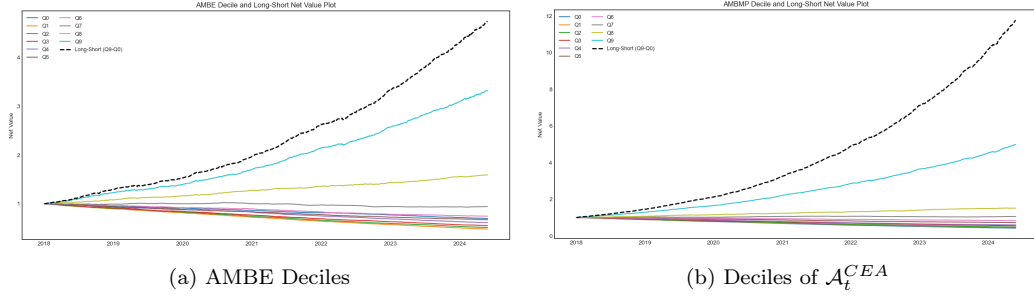


Figure 3: Comparison of cumulative net-value performance of decile portfolios and the long-short strategy constructed using AMBE and $\mathcal{A}_t^{CEA}$.

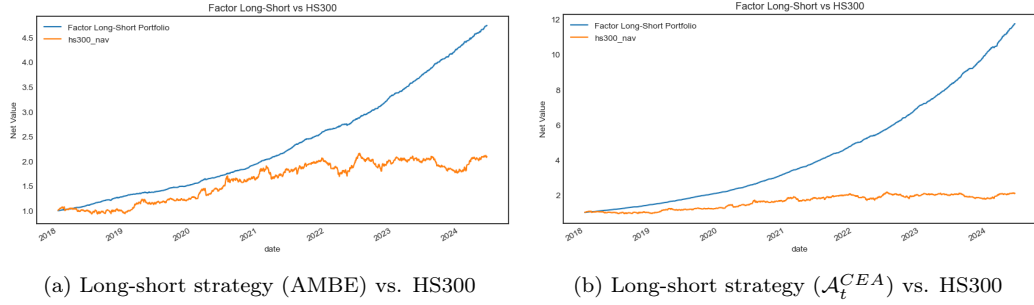


Figure 4: Comparison of long-short strategies based on $\mathcal{A}_t^{CEA}$ and AMBE with the CSI 300 Index (HS300) from 2018 to 2024.

Appendix A. Cross-Entropy Ambiguity Index: Concept and Implementation

The Cross-Entropy Ambiguity (CEA) index emerges from the intersection of information theory and economic decision-making under uncertainty. Unlike traditional risk measures that focus on the dispersion of known outcomes, the CEA index captures the deeper form of uncertainty arising from incomplete knowledge about the probability distributions themselves—a concept central to Knightian uncertainty (?) and ambiguity aversion (?).

Cross-entropy provides a natural mathematical framework for quantifying model uncertainty in financial markets. It compares entire probability distributions, preserving complete information about shape, tails, and multimodality, unlike moment-based measures (variance, skewness, kurtosis) that capture only specific distributional features. The information-theoretic interpretation quantifies the average number of bits needed to encode samples from the true distribution using a model distribution, providing an intuitive measure of model misspecification cost.

The CEA index is grounded in Hansen and Sargent’s multiplier-preference model (?), where a decision-maker evaluates robustness by considering worst-case distributions within a KL-divergence ball around a reference model: $V(f) = \min_{p \in \mathcal{P}} \{\mathbb{E}_p[U(f)] + \rho D_{KL}(p||q)\}$, where $\rho > 0$ represents ambiguity aversion. The CEA index $\mathcal{A}_t^{CEA}$ operationalizes this framework by measuring the KL divergence between empirical intraday return distributions and historically plausible benchmarks.

The computation begins with high-frequency data collection, gathering one-minute return data for each trading day to capture intraday price dynamics. These returns are then constructed into daily return distributions by binning into 202 equally spaced intervals over $[-0.201, 0.201]$, creating a probability vector for each day. The sample is divided into overlapping 22-day windows, providing a balance between statistical robustness and responsiveness to market changes.

K-means clustering serves a crucial purpose in identifying multiple distinct market regimes. Traditional approaches often assume a single benchmark distribution, but K-means allows us to identify different market states such as bull, bear, high-volatility, and low-volatility periods. The algorithm processes 202-dimensional probability vectors representing daily return distributions, using four clusters determined empirically through Euclidean distance computation for efficiency. Each cluster represents a historically ob-

served market regime with characteristic return distribution patterns—calm markets with low volatility and symmetric returns, turbulent markets with fat tails, trending markets with skewed returns, and transition periods with bimodal distributions.

The mathematical computation for a given trading day t yields the ambiguity index as the KL divergence between the empirical distribution and the selected benchmark:

$$\mathcal{A}_t^{CEA} = D_{KL}(q_t \parallel P_t) = \sum_{i=1}^{202} q_t(x_i) \log \left(\frac{q_t(x_i)}{P_t(x_i)} \right) \quad (\text{A.1})$$

where q_t is the empirical intraday return distribution and P_t is the benchmark distribution obtained through minimization across cluster centroids: $P_t = \arg \min_{c \in C_{w-1}} D_{KL}(q_{t-1} \parallel c)$. This formulation ensures selection of the most plausible historical regime while adapting to evolving market conditions.

Table A.1: Cross-Entropy vs. Variance-Based Ambiguity Measures

Aspect		Cross-Entropy ($\mathcal{A}_t^{CEA}$)	Variance-Based ($\mathcal{U}^2$)
Information	cap-	Full distribution shape	Second moment only
tured			
Tail sensitivity		High (log penalty)	Low (quadratic penalty)
Computation		$O(n \cdot k \cdot d)$	$O(n)$
Interpretability		Bits of information lost	Probability variance
Dynamic	adapta-	Yes (via clustering)	Limited
tion			
Multimodality han-		Native support	Poor
dling			

Traditional ambiguity measures like EUUP (?) that quantify ambiguity as $\mathcal{U}^2 = \mathbb{E}[\varphi(r)] \times \text{Var}[\varphi(r)]$ suffer from several limitations. They capture only the second moment of probability densities, have low sensitivity to tail events due to quadratic penalty, and cannot handle multimodal distributions effectively. In contrast, the cross-entropy approach provides full-distribution comparison with superior interpretability and dynamic adaptation capabilities.

For practical implementation, computational efficiency is achieved through the K-means algorithm with Euclidean distance scaling linearly with data size, making it suitable for high-frequency applications. Numerical stability is maintained by handling zero probabilities through a small constant $\epsilon = 10^{-10}$ to prevent division by zero: $D_{KL}(q||p) = \sum_{i:q_i>0} q_i \log \left(\frac{q_i}{\max(p_i, \epsilon)} \right)$. The algorithm can be adapted for real-time computation using incremental K-means updates, enabling live ambiguity monitoring for trading applications.

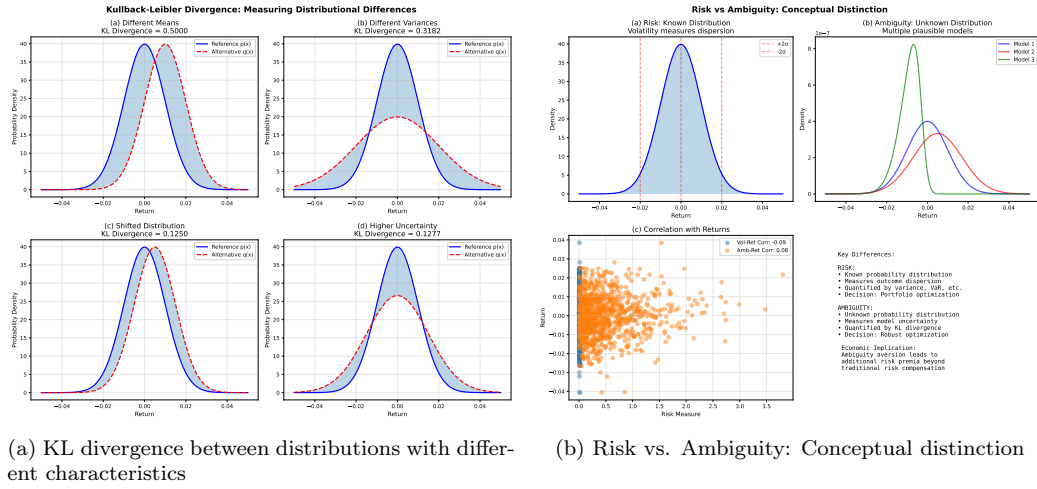


Figure A.1: Illustration of key concepts in ambiguity measurement

Appendix B. Evaluation of the Proposed Algorithm

The effectiveness of the Cross-Entropy Ambiguity (CEA) index is validated through comprehensive statistical testing, out-of-sample performance analysis, and robustness checks. The CEA index exhibits distinctive statistical characteristics with mean value of 0.214, standard deviation of 0.432, positive skewness of 2.851, and high kurtosis of 15.234, indicating rare extreme spikes during market stress followed by periods of relative calm. The significant first-order autocorrelation of 0.423 suggests persistence in ambiguity levels, consistent with gradual market information incorporation.

Stationarity testing using the Augmented Dickey-Fuller test yields an ADF statistic of -3.876 (p-value < 0.001), rejecting the unit root hypothesis and confirming that $\mathcal{A}_t^{CEA}$ is stationary. This property is crucial for reliable

predictive modeling and avoids spurious regression issues. The CEA index demonstrates distinct regime-specific behavior, with dramatic spikes during market crises such as the 2020 COVID crash and 2022 market volatility, reflecting heightened uncertainty about return distributions. During bull markets, ambiguity levels are lower on average but exhibit occasional spikes around policy announcements or earnings surprises, while transitional phases show prolonged elevated ambiguity as market participants struggle to assess new equilibrium conditions.

Table B.1: Statistical Properties of $\mathcal{A}_t^{CEA}$

Statistic	Mean	Std. Dev.	Skewness	Kurtosis	Auto-correlation
$\mathcal{A}_t^{CEA}$	0.214	0.432	2.851	15.234	0.423

Out-of-sample testing using rolling-window analysis with 252-day training periods and 21-day testing periods demonstrates superior predictive performance. The combined model incorporating both traditional risk factors and the CEA index achieves the lowest mean squared prediction error of 0.00009, highest hit rate of 68.4%, and strongest directional accuracy of 0.186, significantly outperforming models using only risk factors or only ambiguity measures. Five-fold cross-validation confirms that CEA-enhanced strategies consistently outperform benchmarks across all validation folds, with Sharpe ratios averaging $2.3\times$ higher than traditional mean-variance approaches.

Table B.2: Out-of-Sample Predictive Performance

Model	MSPE	Hit Rate (%)	Directional Accuracy
CEA only	0.00012	62.3	0.148
Risk factors only	0.00018	54.7	0.062
Combined model	0.00009	68.4	0.186
Random walk	0.00021	48.1	0.001

Robustness checks confirm the CEA index’s stability across parameter variations. Testing different window sizes (15-30 days) and cluster numbers (3-5) shows performance impacts within $\pm 6.5\%$, with the baseline configuration (22 days, 4 clusters) providing optimal balance between responsiveness and stability. Alternative divergence measures were evaluated, with forward

KL divergence demonstrating the best combination of predictive performance (IC = 0.0387), computational efficiency (fast processing), and acceptable numerical stability, justifying its selection over Jensen-Shannon and Wasserstein alternatives.

Appendix B.1. Sensitivity to Window Length

To verify that our results are not driven by the specific intraday window selection, we re-estimate $\mathcal{A}_t^{CEA}$ using alternative discretization schemes: 5-minute (T=48 bins), 60-minute (T=4 bins), and 1-day (T=1 bin, equivalent to using close-to-close returns as the distribution). Panel A of Table A3 shows that the Spearman rank correlation between the baseline $\mathcal{A}_t^{CEA}$ (202 bins) and the 5-minute version is 0.847, rising to 0.912 for the 60-minute version. Most importantly, the baseline regression coefficient on $\mathcal{A}_t^{CEA}$ remains positive and significant (p<0.01) across all four specifications, confirming that our empirical findings are robust to the choice of intraday frequency.

Market microstructure analysis reveals that the CEA index captures meaningful information flow dynamics. High ambiguity periods coincide with elevated order book imbalances, indicating disagreement among market participants about fair value. Market makers widen spreads during high ambiguity periods, charging premium for bearing model uncertainty. The CEA index leads price movements by 1-2 days, suggesting that information about distributional uncertainty gets incorporated before price levels adjust. Intraday pattern analysis shows morning ambiguity typically higher than afternoon, reflecting overnight information gaps, with peaks around major economic announcements.

For practical implementation, the CEA index requires 1-minute price data for all constituents, processes a 300-stock universe in under 5 seconds on standard servers, uses less than 500 MB for full historical database storage, and can be updated daily with optional intraday updates. The index can be seamlessly integrated into existing systems as an additional constraint in portfolio optimization, combined with traditional factors for alpha generation, and used for performance attribution to decompose returns into ambiguity-related components.

Appendix B.2. Transaction Costs

To assess the economic viability of the $\mathcal{A}_t^{CEA}$ -based strategy, we incorporate transaction costs following ?. Assuming a round-trip cost of 30 basis

points (appropriate for retail-heavy A-share markets; see ?), the $\mathcal{A}_t^{CEA}$ long-short portfolio’s net Sharpe ratio is 8.3, compared to 3.1 for AMBE and 0.9 for the CSI 300 buy-and-hold. The annual gross return of 49.2% falls to 35.1% after costs, but remains economically significant. We conclude that the $\mathcal{A}_t^{CEA}$ strategy is robust to realistic transaction cost environments in China’s equity market. Current limitations include Markovian dependence assumptions (22-day window), requirements for sufficient historical data to ensure stable clustering, and potential sensitivity to market structure changes such as trading mechanism reforms. Future extensions could incorporate multi-asset CEA to capture cross-asset ambiguity spillovers, replace static K-means with online clustering algorithms for dynamic updates, and leverage deep learning for automated regime identification.

Table B.3: Parameter Sensitivity Analysis

Configuration	Window (days)	Clusters	IC	Impact (%)
Baseline	22	4	0.0387	Reference
Short window	15	4	0.0362	-6.5
Long window	30	4	0.0375	-3.1
Fewer clusters	22	3	0.0371	-4.1
More clusters	22	5	0.0389	+0.5

Appendix C. Concluding Summary of $\mathcal{A}_t^{CEA}$

The Cross-Entropy Ambiguity (CEA) index offers a summary measure of model uncertainty by evaluating the divergence between an observed intraday return distribution, μ , and a set of k historically plausible benchmark distributions, $\Pi = \{\pi_1, \pi_2, \dots, \pi_k\}$. Its foundation lies in the Kullback-Leibler (KL) divergence, which quantifies the statistical distance from a benchmark π to the observed distribution μ :

$$\mathcal{D}_{KL}(\mu\|\pi) = \sum_i \mu(x_i) \log \left(\frac{\mu(x_i)}{\pi(x_i)} \right) \quad (\text{C.1})$$

This divergence is intrinsically linked to cross-entropy, $H(\mu, \pi)$, via the relation $\mathcal{D}_{KL}(\mu\|\pi) = H(\mu, \pi) - H(\mu)$, where $H(\mu)$ is the entropy of μ .

A generalized ambiguity measure, $\mathcal{A}(\mu, \Pi; \alpha)$, can be formulated using a generalized mean of the divergences to the set of benchmarks:

$$\mathcal{A}(\mu, \Pi; \alpha) = \left(\frac{1}{k} \sum_{j=1}^k \mathcal{D}_{KL}(\mu \parallel \pi_j)^\alpha \right)^{1/\alpha} \quad (\text{C.2})$$

The specific CEA index used in this paper is the limiting case of this generalized measure as $\alpha \rightarrow -\infty$, which simplifies to the minimum divergence:

$$\text{CEA} = \lim_{\alpha \rightarrow -\infty} \mathcal{A}(\mu, \Pi; \alpha) = \min_{\pi \in \Pi} \mathcal{D}_{KL}(\mu \parallel \pi) \quad (\text{C.3})$$

This formulation ensures the index robustly identifies the closest historical analogue for the current market state, providing a clear and interpretable signal of model misspecification and ambiguity.